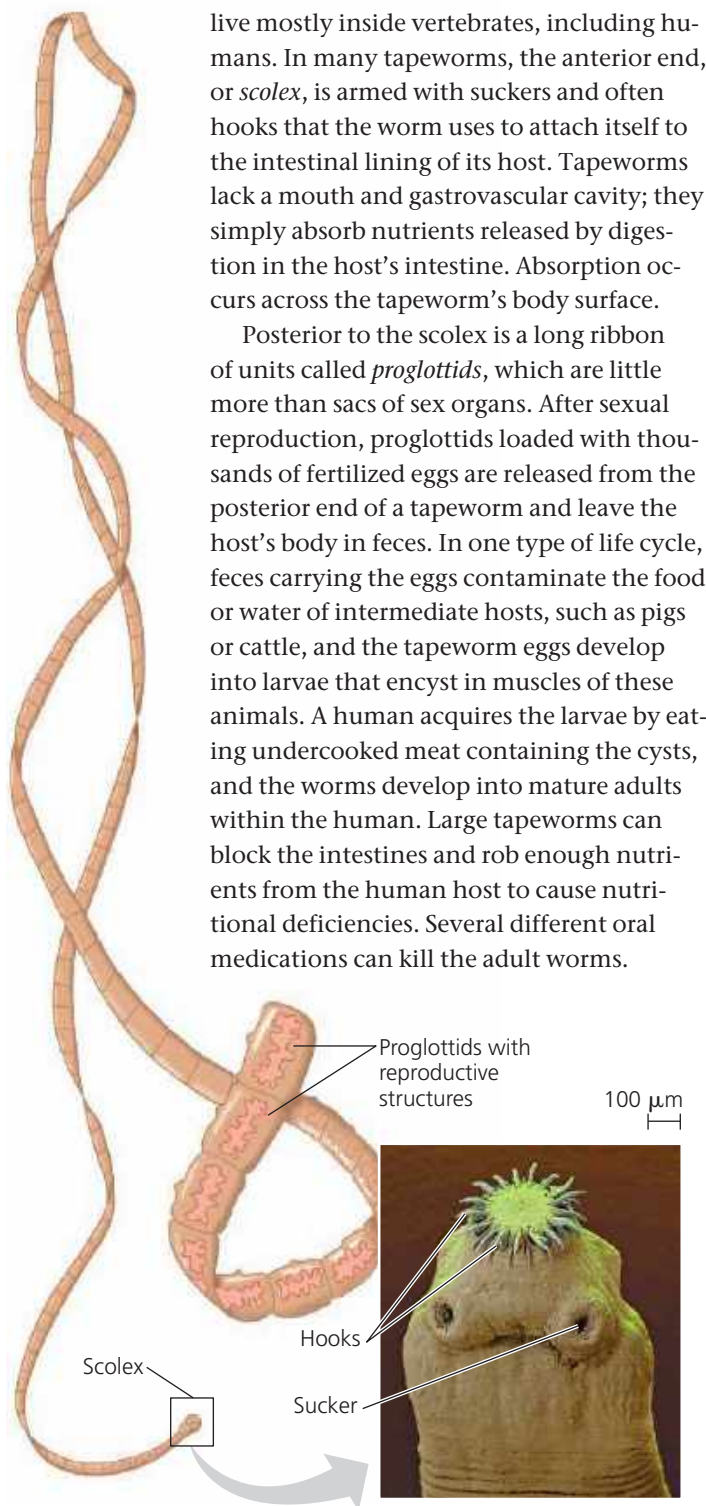




▲ **Figure 33.11 Anatomy of a tapeworm.** The inset shows a close-up of the scolex (colorized SEM).

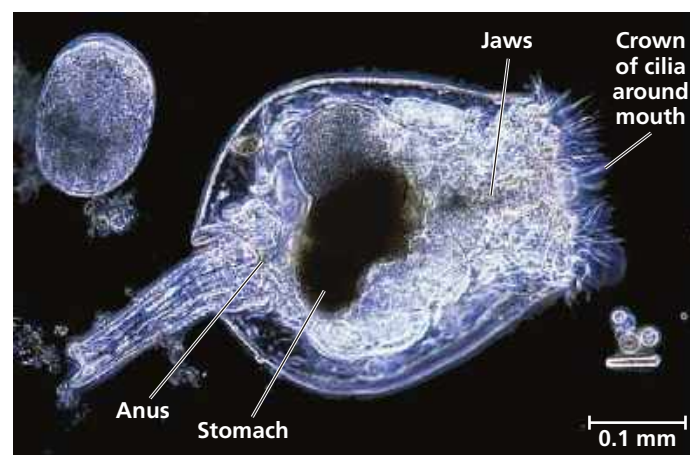
Rotifers and Acanthocephalans

Recent phylogenetic analyses have shown that two traditional animal phyla, the rotifers (former phylum Rotifera) and the acanthocephalans (former phylum Acanthocephala), should be combined into a single phylum, Syndermata. Each of the two groups has distinctive characteristics.

live mostly inside vertebrates, including humans. In many tapeworms, the anterior end, or *scolex*, is armed with suckers and often hooks that the worm uses to attach itself to the intestinal lining of its host. Tapeworms lack a mouth and gastrovascular cavity; they simply absorb nutrients released by digestion in the host's intestine. Absorption occurs across the tapeworm's body surface.

Posterior to the scolex is a long ribbon of units called *proglottids*, which are little more than sacs of sex organs. After sexual reproduction, proglottids loaded with thousands of fertilized eggs are released from the posterior end of a tapeworm and leave the host's body in feces. In one type of life cycle, feces carrying the eggs contaminate the food or water of intermediate hosts, such as pigs or cattle, and the tapeworm eggs develop into larvae that encyst in muscles of these animals. A human acquires the larvae by eating undercooked meat containing the cysts, and the worms develop into mature adults within the human. Large tapeworms can block the intestines and rob enough nutrients from the human host to cause nutritional deficiencies. Several different oral medications can kill the adult worms.

▼ **Figure 33.12 A rotifer.** Although smaller than many unicellular protists, rotifers are multicellular and have specialized organ systems (LM).



Rotifers

There are roughly 1,800 species of rotifers, tiny animals that inhabit freshwater, marine, and damp soil habitats. Ranging in size from about 50 µm to 2 mm, rotifers are smaller than many protists but nevertheless are multicellular and have specialized organ systems (**Figure 33.12**). In contrast to cnidarians and flatworms, which have a gastrovascular cavity, rotifers have an **alimentary canal**, a digestive tube with two openings, a mouth and an anus. Internal organs lie within the *hemocoel* (see Figure 32.9b). Fluid in the hemocoel serves as a hydrostatic skeleton. Movement of a rotifer's body distributes the fluid throughout the body, circulating nutrients.

The word *rotifer* is derived from the Latin meaning "wheel-bearer," a reference to the crown of cilia that draws a vortex of water into the mouth. Posterior to the mouth, rotifers have jaws called trophi that grind up food, mostly microorganisms suspended in the water. Digestion is then completed farther along the alimentary canal. Most other bilaterians also have an alimentary canal, which enables the stepwise digestion of a wide range of food particles.

Rotifers exhibit some unusual forms of reproduction. Some species consist only of females that produce more females from unfertilized eggs, a type of asexual reproduction called **parthenogenesis**. Some other invertebrates (for example, aphids and some bees) and even some vertebrates (for example, some lizards and some fishes) can also reproduce in this way. In addition to being able to produce females by parthenogenesis, some rotifers can also reproduce sexually under certain conditions, such as high levels of crowding. The resulting embryos can remain dormant for years. Once they break dormancy, the embryos develop into another generation of females that reproduce asexually.

It is puzzling that many rotifer species persist without males. The vast majority of animals and plants reproduce sexually at least some of the time, and sexual